William Tun

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Education

Johns Hopkins University – B.S. in Mechanical Engineering August 2025 - May 2029

Relevant coursework: Autodesk Inventor, Electrical Systems, DE Statistics, Mechanics I & II, Calculus I, II, & III, MechE & Design (SolidWorks), MechE Lab, Statics and Structures.

Skills

Programming Languages: C++, Python

Software/Tools: MATLAB, SolidWorks, Autodesk Inventor, Fusion 360, Onshape, CATIA, Creo, ANSYS, Arduino, Excel, PowerPoint, Word, CNC machining, manual lathe, manual mill, TIG/MIG Welding, GD&T, SolidWorks PDM.

Experiences

JHU WSE Manufacturing, Johns Hopkins **-** Baltimore, MD **October 2025 – Present**

Manufacturing Lab Technician

- Maintain milling machines, lathes, EDM equipment, 3D printers, and welding equipment while helping university-affiliated users bring concepts to life through engineering drawings, fabrication planning, and manufacturing custom components from 6061 aluminum and steel.
- Design components for proper fitment within existing assemblies, including most of the Baja SAE vehicle's control arm bushings and additional lathe-machined parts, accounting for mating dimensions, clearances, and manufacturing constraints to support assembly and operation.
- Develop custom jigs and fixtures for testing and validation projects, translating project requirements into manufacturable designs that position and secure components during testing; assist with prototyping, assembly, and design revisions to meet user needs.

Mechanical Engineering Intern, Joint Base Andrews – Oxon Hill, MD **May 2025 – August 2025**

- Designed a 3D printing filament recycling machine in SolidWorks, capable of processing engineering-grade and high-temperature polymers (PA6-CF, PETG, Nylon, ABS, PC, and PEEK).
- Developed ANSYS simulations to model conductive heat transfer across components and iterated design improvements based on analysis and feedback.
- Recycled filament maintained the original material properties of pre-recycled filament, cutting annual filament costs by 20%.

Projects

Front Upright Design & Manufacturing, Blue Jay Racing **May 2026 – August 2026**

- Designed and modeled the lightest front upright in team history using SolidWorks, optimizing geometry to minimize mass while maintaining structural integrity and meeting all Load cases.
- Collected real-world suspension loading data from Baja SAE competition events and endurance testing, then developed an ANSYS finite element model to replicate braking, cornering, impact, and landing load cases for design validation and optimization.
- Created detailed manufacturing drawings with GD&T for the final front upright and collaborated with a 5-axis CNC machining sponsor to manufacture the component from a solid billet of 7075-T6 aluminum, ensuring critical tolerances and bearing fits were met.
- Fabricated and assembled filament recycler components using resin and FDM printing, incorporating design features to support material feeding, heating, and extrusion.

Enclosure & Lower Frame Design lead, Oxon Hill Clipper racers **August 2024 – May 2025**

- Designed and cast an aluminum lower vehicle frame to house and protect the battery system, modeled in SolidWorks and fabricated using A356 aluminum for its high strength-to-weight ratio and corrosion resistance, ensuring secure battery mounting under dynamic loads.
- Performed ANSYS simulations on multiple bolt and mounting-point configurations to evaluate stress concentrations, fastener loading, and frame deflection under acceleration, braking, and cornering loads.
- Reduced vehicle mass by 32.5 lb, decreasing inertia and load transfer to improve transient response, cornering stability, and predictability under aggressive maneuvering.